TOSHA ENGINEERING PRIVATE LIMITED
14/3, Mile Stone, 1st Floor, Sector-27 B, Main Mathura Road, Behind Vijay Sales, Faridabad (HR)-121 003, India
Contact No. : 7986427191, 9560066068 / 70
E-mail : info@toshaengg.com, cs@toshaengg.com
GSTIN : 06AAICT8111E1ZJ

Quotation No : Q24000310

Date : 28-May-2024

Tata International Vehicle Applications Private Limited
Plot No. E-1, RIICO Industrial Area,Jaipur-Bhilwara Highway, Shrinagar, Ajmer
KISHANGARH
RAJASTHAN
305025

Kind Attention : Mr./Ms. Manish Singodia

Dear Sir,

Our team understands that selecting gas generation systems is a difficult task.
We have the expertise to help you being a prime manufacturer and supplier of gas generation systems.

Absstem has the resources, facilities, technology, and expertise required to ensure a successful project.
Our approach is low-risk because of our previous experience with large and complex projects.
Post-installation, our shield service division, with diligent accuracy and functionality, oversees a team of technicians who provide warranty and post-warranty support to your newly-installed gas generation systems.
Our customers have relied on scale up their production.

Our products have consistently delivered absolute purity requirements.
The industry's lowest ever production downtime.
Reduce OPEX by at least 70%.

We wish to form a long-term relationship with the goal to develop the most cost-effective and comprehensive solutions for your needs.
With an experienced team of professional personnel already in place, our team is prepared to move swiftly to ensure that we meet critical objectives & timelines.
Please feel free to contact in case of any query.

Looking forward for a positive response.

Thanks and Regards,

For TOSHA ENGINEERING PRIVATE LIMITED

Rajesh Ashok

9910661209

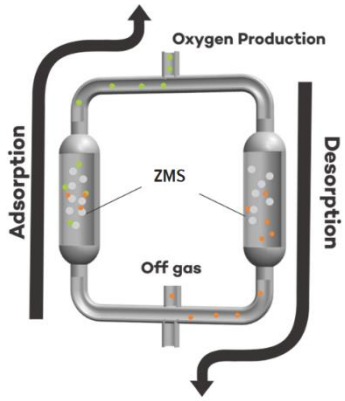
www.toshaengg.com

idos customs

Industrial PSA Oxygen Generator



PSA Technology



PSA Process explaining oxygen production

Industrial Oxygen Generator mainly consists of two adsorbers filled with zeolite molecular sieves. Pressurized air is passed through one of these adsorbers, zeolite having a property to attract nitrogen, adsorbs Nitrogen from air and oxygen is allowed to pass through and move into the storage tank as the final product. As the adsorber approaches saturation, the air is allowed to pass through the other adsorber while this adsorber is being regenerated to ensure a continuous supply of Oxygen. This changeover takes place through a specially designed microcontroller to ensure a hassle-free operation. The quality of oxygen produced is monitored by an Oxygen analyser.

Key Features

- Shipped ready for installation
- 5 minutes to run and never have to stop it!
- Easy installation
- Low operating expenses compared to gas cylinders/liquid Oxygen
- Maintenance costs are very less
- Embedded Safety Interlocks
- Space-saving design
- Eliminates all hazards which are there with gas cylinder or liquid Oxygen
- Longevity: at least 10 years

absstem REINVENTING PERFORMANCE

Absstem Industrial Solutions Pvt. Ltd.
Plot No. 591, Sector 8, Phase IV
IMT Manesar, Gurugram, Haryana - 122051
Contact - 1800 3010 3394 Email - indsales@absstem.com
CIN: U29256HR2022PTC103387

Description

a.	Product Name	Oxygen Gas Generator
b.	Model Range	**IDOS C**
c.	Technology	PSA (Pressure Swing Adsorption)

Specifications

a.	Oxygen Flow Rate	As per Performance Sheet
b.	Oxygen Purity	As per Performance Sheet
c.	Oxygen Delivery Pressure	5.5 Barg (Higher Pressure Options Available)
d.	Ambient conditions	
	Temperature	0 to 50 Deg. Celsius
	Relative Humidity	up to 80 % (Max.)
	Altitude	up to 1000 m above Sea level

Scheme Offered – (PSA Only)

This scheme will comprise of

- Air package with inlet air filtration system
  (Air compressor, Air dryer, Water Separator, Oil filters)
- Industrial Oxygen generator
- Storage Tank
- Oxygen Analyzer

Unique Features of Absstem

a. All the fabricated items are cleaned and degreased following our strict quality policy.
b. All the Instruments are specially manufactured to make them suitable for Oxygen gas.
c. Guaranteed 10+ years of healthy life.
d. We provide the purest **Oxygen** in Industry.
e. Data logging and Graphical Reports for all parameters available.
f. Test certificates of all instruments are available on demand.

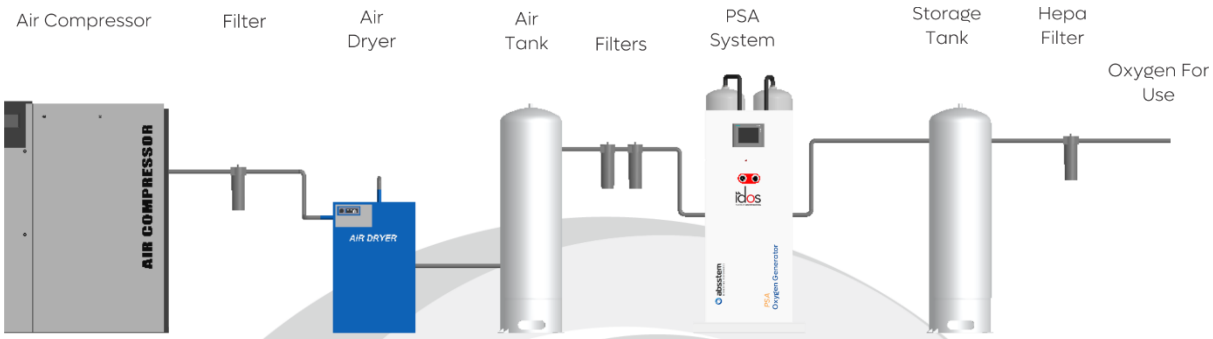
Onsite Gas Generation vs Liquid Tank/Cylinder Alternative

Non-stop availability: 24 hours a day, 7 days a week
Easy integration within existing installations
Your own independent supply of industrial gas
Significant economies for scale and lower operational costs: no rental charges, logistical expenses and bulk user evaporation losses
No safety hazards when handling high-pressure cylinders

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Equipment Description

The Industrial Oxygen Generator consists of

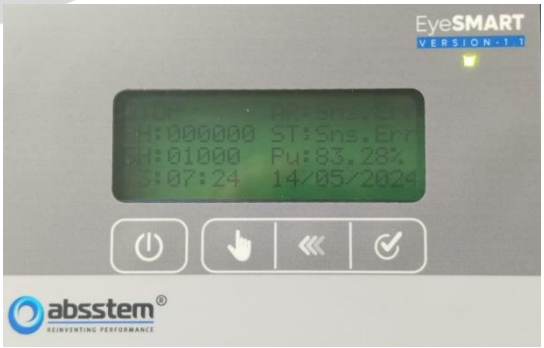


Air Compressor	An Air compressor is used to compress the atmospheric air to a pressure required for the separation of **Oxygen** from the air.
Refrigeration dryer	Compressed air then passes through a refrigerated dryer to remove moisture (which is harmful to sieves) from the air.
Air Tank	Dry air is collected in an air tank to remove the pulsations which serve the dynamic (changing) requirement of air by the PSA system.
Multistage Filtration system	Before air enters the PSA system, it is passed through multistage filtration removing moisture, particulates and oil content to the specified levels.
PSA system	Pressurized air is passed through PSA Towers filled with Molecular sieves to separate **Oxygen** from Air.
Storage Tank	Industrial Grade **Oxygen** produced from IDOS C is stored in this tank to supply a continuous flow of **Oxygen**.
Oxygen Analyzer	Oxygen produced is monitored by an **Oxygen** analyser. In case, O2 purity goes below the specified level, it gives an alarm and switches it to vent mode.

EYE-SMART Touch Screen Display

(User-friendly touchscreen interface for easy operation and maintenance)

Absstem industrial Oxygen generator complies with all industry standards and regulation



- Alarm management system
- Plug and play
- Fully automatic
- Process data display
- Consumables running hours display
- Data logging via USB drive
- Graphical representation

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Additional upgrades

- Digital flow meter with totaliser
- High-pressure Oxygen tank
- Cylinder manifold and alarm management system
- Remote monitoring

Plant Operation

- Unit operates 24-hrs a day, 7-days a week reliably, automatically & safely

Automation

- The Plant is fully automatic and needs only electrical power for running the Air compressor
- The machine automatically enters into standby mode when the level in the Storage Tank reaches its set limit. This helps to increase the life of the Air compressor.

International Quality standards

A strict quality procedure, following the leading international pharmacopoeia and Industrial directives, is followed in the selection of all mechanical and electrical instruments, vessel fabrication and designing of Industrial Oxygen generators.

Itemised Supply description

No.	Item description	Make	Qty
a	Air Compressor with Filters	ABSSTEM APPROVED	Client Scope / Quoted Separately
b	Refrigeration Dryer	ABSSTEM APPROVED	Client Scope / Quoted Separately
c	Air Tank	ABSSTEM	Client Scope / Quoted Separately
d	Set of Filters	ABSSTEM	1 no.
e	PSA unit (IDOS C)	ABSSTEM	1 no.
f	Electrical Panel	ABSSTEM	1 no.
f	Digital Oxygen Analyzer	ABSSTEM	1 no.
g	Storage Tank	ABSSTEM	1 no.

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Electrical Protection

The client should make proper arrangements for earthing/grounding systems for the Industrial **Oxygen** Generator at their facility.

Enclosure Panel

- Ergonomically designed metal enclosure with powder-coated finish for longer life.
- High-strength IP-rated panel fittings from Germany for a robust enclosure.
- Fan filters from **Rittal** are IP-55 rated making the electrical enclosure's internal environment dust-free.
- Easy connectivity to any place of installation with req. port connections.
- High-strength base frame designed for higher load and easy transportation.
- Vessels are designed per ASME section VIII DIV I. Vessels are epoxy paint coated which is considered good for longevity and corrosion resistance.
- All electrical items are CE certified with operating voltage at 24 VDC

Annual Maintenance Contract (AMC)

Our in-house support Absstem shield is 24*7 open and we assure a timely service maintenance schedule. Per the schedule mentioned in the operational manual, we suggest the operator change the filters by following a simple protocol.

It is very easy to change or replace as our commissioning engineer plus our commissioning engineer and tech support office will always guide you about the activity.

Best ROI Guaranteed

ROI within 6 to 18 months when compared to conventional methods

Generator performance depends on the component selection. Our processes make this job easier as all equipment is selected based on more than 3 decades of research. IDOS C is a fully automatic (operator-free operation) that ensures peace of mind.

We deliver what we commit!!

Thank You

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Product Description

Product	Description	Make	Qty
Absstem Air Package	69 CFM ABSSTEM COMPRESSOR PACKAGE		1

Caption	Specification
Air Pressure	8 Barg
Flow in CFM	69
Air Dryer	Included
Air Filter	Included
Air Vessel in Liters	500
Power in KW	11
Make	Absstem Approved Brand

Product	Description	Make	Qty
IDOS C	IDOS 7C Oxygen		1

Caption	Specification
Gas Generated	Oxygen
Flow	7 Nm3/Hr
Purity	95 Percent
Compressor	In Your Scope / Quoted Separately
Dryer	In Your Scope / Part of Compressor Package
Air Vessel	In Your Scope / Part of Compressor Package
Dry Air Required	58 CFM
Power Required	230V 50 Hz Single Phase
Module Option	PSA Plant Only

Product	Description	Make	Qty
IDOS Gas Storage	IDOS 07 ST Oxygen		1

Caption	Specification
Water Volume	1500 Litres
Gas Stored	Oxygen
Gas Stored Volume	8.25 M3
Working Pressure	5.5 Barg
Hydrotest Pressure	9 Barg

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Commercial Offer

S No	Item Code	Item Name	Description	HSN	Qty	Unit Price	GST(%)	Amount
1	KS-15-P	69 CFM	Absstem Air Package	84148090	1	5,58,000.00	18.00	5,58,000.00
2	IDOS 7C	IDOS 7C	IDOS C	842139	1	10,96,800.00	18.00	10,96,800.00
3	IDOS 07 ST	IDOS 07 ST	IDOS Gas Storage	842139	1	1,56,000.00	18.00	1,56,000.00
							Sub Total	18,10,800.00
							GST(%) @18.00	3,25,944.00
							Grand Total	21,36,744.00

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Terms And Conditions

1.	EXCLUSIONS & CLIENTS SCOPE	All erection / civil work, All Power Cables & Cabling out of battery limit of PSA Plant, Cable Trenches etc. Unloading / Shifting of Gas Plant at Clients site will be in Client Scope. Additional gas piping, cooling water (if required).
2.	SUPERVISION OF ERECTION & COMMISSIONING	There shall be no commissioning charges, however you shall arrange freeTo & Fro Air Ticket, lodging / boarding, and local conveyance for our Engineers during supervision of Installation of Gas Plant at site on free of cost basis. Our Engineers will be there for Supervision only. Any manpower / fitter required for installation work shall be provided by you.
3.	P&F	3% Extra.
4.	Freight	Extra at Actuals.
5.	Insurance	In your Scope.
6.	Payment Terms	50% Advance with your Prestegious Purchase Order and Balance against PI.
7.	Delivery	6-8 Weeks from the PO and Advance. Rest can be confirmed for Availability.
8.	Warranty	We hereby confirm that, our equipment shall have Suppliers warranty for a period of 12 months from the date of Commissioning or 18 months from date of Dispatch whichever is earlier.
9.	GST	As per the Commercial Table.
10.	Remarks	.

Hope you will find our offer in line with your requirement.

Should you require any further clarification/information, please feel free to contact us.

Thanking you, and assuring you of our best attention at all times.

Very Yours truly,

For TOSHA ENGINEERING PRIVATE LIMITED

Rajesh Ashok

9910661209

www.toshaengg.com

Cost Comparison (PSA Vs Cylinder)

	units	Cost Comparison
Daily Consumption of Oxygen Cylinders	Nos	24
Total Consumption of Oxygen Cylinder per month	Nos	624
Cost of Oxygen Cylinder	Rupees	133
Monthly Cost of Oxygen	Rupees	82992
Per M3 Oxygen Cost in Cylinder	Rupees	22.2
Plant Running Time	hr.	24.0
Per hour consumption	Nm3/hr	7.0
PSA Plant Flow	NM3/hr	7.5
Compressor kw.	KW	9
Power	KW	8.25
Power Cost / Unit	Rupees	11
Unit Price / M3 in with PSA Oxygen	Rupees	12.1
Saving/m3	Rupees	10.1
Monthly cost of Oxygen with PSA plant	Rupees	45302.40
Monthly saving by using PSA	Rupees	37689.60
Yearly saving by using PSA	Rupees	452275.20
Final Saving in PSA Oxygen Plant	Rupees	452275.20
Plant cost (Budgetary Price)	**Rs.**	**2782800.00**
ROI	(In Years)	6.15

Exclusions

Pipeline from Plant to Point of Application